Ordered groups, arbitrary length

B Batch **I** Left free **K** Contract **J** Right free

K = $[k_0, \ldots, k_{n-1}]$, with arbitrary length n.

dot(X, Y), one concrete binding

X	b_a	i0	k0	k1	i1	
	0	1	2	3	4	
Y	b_b	j0	k1	j1	j2	k0
	0	1	2	3	4	5

b_a ↔ b_b share batch group B.

Group	X positions	Y positions
B	[0]	[0]
I	[1, 4]	–
K	**[2, 3]**	**[5, 2]**
J	–	[1, 3, 4]

K pairs: X[2] ↔ Y[5], then X[3] ↔ Y[2].

Output: B ++ I ++ J

Reduced over K

b	i0	i1	j0	j1	j2